

Complex Eigenvalues and the Trace–Determinant Plane

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Definition: Linear spiral

The origin of $\dot{\mathbf{x}} = A\mathbf{x}$, with $A \in \mathbb{R}^{2 \times 2}$, is a **spiral** if A has nonreal eigenvalues with nonzero real part.

Definition: Linear center

The origin of $\dot{\mathbf{x}} = A\mathbf{x}$, with $A \in \mathbb{R}^{2 \times 2}$, is a **center** if every nonzero trajectory is a closed curve surrounding the origin.

Question: What do complex eigenvalues mean in a real system?

Let $\alpha \in \mathbb{R}$, $\beta \in \mathbb{R} \setminus \{0\}$, and

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Find the eigenvalues and the polar equations.

Solution. The characteristic polynomial is

$$(\alpha - \lambda)^2 + \beta^2,$$

so

$$\lambda = \alpha \pm i\beta.$$

Write $x = r \cos \theta$, $y = r \sin \theta$, with $r > 0$ and a continuous choice of angle θ . Then

$$r\dot{r} = x\dot{x} + y\dot{y} = \alpha(x^2 + y^2) = \alpha r^2,$$

$$\dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2} = \beta.$$

Therefore

$$r(t) = r_0 e^{\alpha t}, \quad \theta(t) = \theta_0 + \beta t.$$

Here $r_0 = r(0) > 0$ and $\theta_0 = \theta(0)$. The origin is an asymptotically stable spiral if $\alpha < 0$, an unstable spiral if $\alpha > 0$, and a center if $\alpha = 0$. The angular revolution time is $2\pi/|\beta|$, and the sign of β gives the orientation. When $\alpha \neq 0$, this is the time for one revolution, not a period of the nonclosed trajectory.

Notation: Trace

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

write

$$\tau = \operatorname{tr} A = a + d.$$

Notation: DeterminantFor $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, write

$$\Delta = \det A = ad - bc.$$

Notation: DiscriminantFor a real 2×2 matrix with trace τ and determinant Δ , write

$$D = \tau^2 - 4\Delta.$$

Definition: Matrix exponentialFor a real 2×2 matrix A and $t \in \mathbb{R}$, the **matrix exponential** is

$$e^{At} := \sum_{k=0}^{\infty} \frac{(At)^k}{k!}.$$

Observation: Matrix exponential and the initial-value problemFor a constant real 2×2 matrix A ,

$$\frac{d}{dt} e^{At} = A e^{At}, \quad e^{A0} = I,$$

where I is the identity matrix. Thus the solution of $\dot{\mathbf{x}} = A\mathbf{x}$, $\mathbf{x}(0) = \mathbf{x}_0$, is

$$\mathbf{x}(t) = e^{At} \mathbf{x}_0.$$

Observation: Eigenvalues from trace and determinant

The characteristic polynomial and eigenvalues are

$$p(\lambda) = \lambda^2 - \tau\lambda + \Delta, \quad \lambda_{1,2} = \frac{\tau \pm \sqrt{D}}{2}.$$

Thus

$$\lambda_1 + \lambda_2 = \tau, \quad \lambda_1 \lambda_2 = \Delta.$$

Theorem: Trace–determinant classification for an isolated origin

Let A be a real 2×2 matrix with $\Delta = \det A \neq 0$. Then the origin is the unique equilibrium of $\dot{\mathbf{x}} = A\mathbf{x}$, and

condition	type at the origin	stability
$\Delta < 0$	saddle	unstable
$\Delta > 0, D > 0$	node	$\tau < 0$: asymptotically stable; $\tau > 0$: unstable
$\Delta > 0, D < 0, \tau \neq 0$	spiral	$\tau < 0$: asymptotically stable; $\tau > 0$: unstable
$\Delta > 0, \tau = 0$	center	neutrally stable
$\Delta > 0, D = 0$	star or degenerate node	$\tau < 0$: asymptotically stable; $\tau > 0$: unstable

Proof. If $\Delta < 0$, the eigenvalues are real with opposite signs. If $\Delta > 0$ and $D > 0$, they are real with the same sign, determined by their sum τ .

For $D < 0$, put $\beta = \sqrt{-D}/2 > 0$ and $B = A - (\tau/2)I$. Direct multiplication gives $B^2 = -\beta^2 I$, so the exponential series gives

$$e^{At} = e^{\tau t/2} \left(I \cos \beta t + \frac{B}{\beta} \sin \beta t \right).$$

For $\mathbf{x}_0 \neq \mathbf{0}$, the vectors $\mathbf{x}_0$ and $B\mathbf{x}_0$ are independent: a real eigenvalue of B would contradict $B^2 = -\beta^2 I$. Thus the parenthesized factor traces an ellipse when applied to $\mathbf{x}_0$; the exponential shrinks or expands it when $\tau \neq 0$. The parenthesized matrix is bounded and periodic. If $\tau < 0$, its exponentially decaying prefactor proves both stability and attraction. If $\tau > 0$, successive times $t = 2\pi k/\beta$ multiply each nonzero initial state by $e^{\tau\pi k/\beta}$ for $k = 1, 2, \dots$, proving instability. If $\tau = 0$, every nonzero solution is periodic and stays close whenever it starts close: the origin is a neutrally stable center. Its period is $2\pi/\beta = 2\pi/\sqrt{\Delta}$.

If $D = 0$ and $\Delta > 0$, the repeated eigenvalue is $\tau/2 \neq 0$; the eigenspace dimension distinguishes a star from a degenerate node.

Proposition: Zero-determinant boundary

Let A be a real 2×2 matrix with $\Delta = \det A = 0$, and let $\tau = \operatorname{tr} A$.

condition	equilibria	stability of the origin
$\tau < 0$	$\ker A$ is a line	Lyapunov stable, not attracting
$\tau > 0$	$\ker A$ is a line	unstable
$\tau = 0, A = 0$	$\mathbb{R}^2$	Lyapunov stable, not attracting
$\tau = 0, A \neq 0$	$\ker A$ is a line	unstable

Proof. If $\tau \neq 0$, the distinct eigenvalues are 0 and τ . The zero eigendirection consists of equilibria, while the other component decays for $\tau < 0$ and grows for $\tau > 0$.

Direct multiplication gives $A^2 - \tau A + \Delta I = 0$. If $\tau = 0$, this gives $A^2 = 0$. When $A = 0$, every solution is constant. When $A \neq 0$,

$$e^{At} = I + tA,$$

so arbitrarily small initial states outside $\ker A$ produce solutions that leave any fixed neighborhood of the origin.

Visualization: Trace–determinant plane

With horizontal coordinate τ and vertical coordinate Δ ,

region or curve	portrait
$\Delta < 0$	saddle
$0 < \Delta < \tau^2/4$	node
$\Delta > \tau^2/4, \tau < 0$	asymptotically stable spiral
$\Delta > \tau^2/4, \tau > 0$	unstable spiral
$\tau = 0, \Delta > 0$	center
$\Delta = \tau^2/4 > 0$	star or degenerate node
$\Delta = 0, \tau < 0$	Lyapunov-stable line of equilibria
$\Delta = 0, \tau > 0$	unstable line of equilibria
$\Delta = \tau = 0$	zero matrix or unstable case

Within the half-plane $\Delta > 0$, the parabola $\Delta = \tau^2/4$ separates real-eigenvalue nodes from complex-eigenvalue portraits; the latter are spirals except for centers on $\tau = 0$. If $\tau < 0$ and $\Delta > 0$, the origin is asymptotically stable; if $\tau < 0$ and $\Delta = 0$, it is Lyapunov stable but not attracting. If $\tau > 0$, the origin is unstable, and every point with $\Delta < 0$ is a saddle.

Question: Can trace and determinant classify without solving?

Classify

$$A_1 = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}.$$

Solution. For A_1 ,

$$\tau = 5, \quad \Delta = 4 - 6 = -2 < 0,$$

so the origin is a saddle.

For A_2 ,

$$\tau = 6, \quad \Delta = 8 - 3 = 5, \quad D = 36 - 20 = 16 > 0.$$

The origin is an unstable node. Its eigenvalues are

$$\lambda = \frac{6 \pm 4}{2} = 5, 1.$$

Question: How do we solve a stable spiral?

For

$$A = \begin{pmatrix} -1 & -4 \\ 1 & -1 \end{pmatrix},$$

classify the origin, determine the orientation, and solve the initial-value problem $\mathbf{x}(0) = (2, 0)$.

Solution.

$$\tau = -2, \quad \Delta = 5, \quad D = -16,$$

so $\lambda = -1 \pm 2i$ and the origin is an asymptotically stable spiral. At $(1, 0)$,

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix},$$

Question: How do we solve a stable spiral? (continued)

so the rotation is counterclockwise.

Let

$$B = A + I = \begin{pmatrix} 0 & -4 \\ 1 & 0 \end{pmatrix}.$$

Since $B^2 = -4I$,

$$e^{At} = e^{-t} \left(I \cos 2t + \frac{B}{2} \sin 2t \right) = e^{-t} \begin{pmatrix} \cos 2t & -2 \sin 2t \\ \frac{1}{2} \sin 2t & \cos 2t \end{pmatrix}.$$

Thus

$$x(t) = 2e^{-t} \cos 2t, \quad y(t) = e^{-t} \sin 2t.$$

One revolution takes time π , and

$$\mathbf{x}(t + \pi) = e^{-\pi} \mathbf{x}(t).$$

Thus the distance from the origin shrinks by $e^{-\pi}$ per turn. The spiral trajectory itself is not periodic.

Question: Why do boundary portraits require more than eigenvalue multiplicity?

Classify

$$A_1 = -I, \quad A_2 = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Solution. Both A_1 and A_2 satisfy

$$\tau = -2, \quad \Delta = 1, \quad D = 0.$$

For A_1 , every nonzero vector is an eigenvector, so the origin is an asymptotically stable star node.

For A_2 ,

$$\ker(A_2 + I) = \text{span}\{(1, 0)\},$$

so the origin is an asymptotically stable degenerate node.

For A_3 , $\Delta = 0$, $\tau = -1$, and

$$x(t) = x_0 e^{-t}, \quad y(t) = y_0.$$

Every point on the y -axis is an equilibrium. The origin is Lyapunov stable but not attracting.

Definition: Linear love-affair model

Let $R(t)$ denote Romeo's feeling toward Juliet and $J(t)$ Juliet's feeling toward Romeo, with positive values representing love and negative values representing hate. For constants $a, b, c, d \in \mathbb{R}$, a linear response model is

$$\dot{R} = aR + bJ, \quad \dot{J} = cR + dJ.$$

This is the **linear love-affair model**.

Observation: Interpretation of the response coefficients

In the linear love-affair model, the diagonal coefficients a, d measure responses to one's own feelings, while the off-diagonal coefficients b, c measure responses to the other person's feelings.

Question: What happens to responsive Romeo and fickle Juliet?

Let

$$\dot{R} = aJ, \quad \dot{J} = -bR, \quad a, b > 0.$$

Classify the origin and find the trajectories and period.

Solution.

$$A = \begin{pmatrix} 0 & a \\ -b & 0 \end{pmatrix}, \quad \tau = 0, \quad \Delta = ab > 0.$$

Thus the origin is a center with eigenvalues $\pm i\sqrt{ab}$. Moreover,

$$H(R, J) = bR^2 + aJ^2$$

is conserved because

$$\dot{H} = 2bR(aJ) + 2aJ(-bR) = 0.$$

For $C > 0$, the level sets $H = C$ are the nonzero trajectories and are ellipses. The level set $H = 0$ is the equilibrium at the origin. Every nonzero trajectory has period

$$T = \frac{2\pi}{\sqrt{ab}}.$$

At $(R, 0)$ with $R > 0$, $\dot{J} = -bR < 0$, so the motion is clockwise.

Question: What happens to two identically cautious lovers?

Let

$$\dot{R} = aR + bJ, \quad \dot{J} = bR + aJ, \quad a < 0 < b.$$

Give the complete classification as b varies.

Solution. The eigenpairs are

$$\lambda_+ = a + b, \quad \mathbf{v}_+ = (1, 1), \quad \lambda_- = a - b, \quad \mathbf{v}_- = (1, -1).$$

Since $a - b < 0$,

condition	type	long-time behavior
$0 < b < -a$	asymptotically stable node	$(R, J) \rightarrow (0, 0)$
$b = -a$	line $R = J$ of equilibria	$(R, J) \rightarrow (c, c)$
$b > -a$	saddle	$W^u = \{R = J\}, W^s = \{R = -J\}$

In the middle case, $c = (R(0) + J(0))/2$.